

### Orthogonal Projection - Revised.

Let  $V$  be an inn. prod space, (not necessary finite dimension!) and  $W \subseteq V$  be a subspace.

For any vector  $\beta \in V$ , if  $\alpha \in W$  satisfies that:

$$\|\beta - \alpha\| \leq \|\beta - \gamma\| \quad \text{for all } \gamma \in W \quad \dots (K)$$

Then, the  $\alpha$  is called best approximation to  $\beta$  by  $W$ , simply b.app.

Thm.  $\alpha \in W$  is b.app to  $\beta \in V$  by  $W$  iff for all  $\mu \in W$ ,  $\langle \beta - \alpha, \mu \rangle = 0$ .

Proof. For fixed  $\alpha \in W$  and  $\beta \in V$ , for any  $\gamma \in W$ ,

$$\|\beta - \gamma\|^2 = \|\beta - \alpha + \alpha - \gamma\|^2 = \|\beta - \alpha\|^2 + \|\alpha - \gamma\|^2 + 2 \cdot \operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle$$

Therefore, the condition (K) holds if and only if

$$0 \leq \|\alpha - \gamma\|^2 + 2 \cdot \operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle \quad \text{for all } \gamma \in W.$$

Now, take  $\gamma = \alpha + \frac{\langle \beta - \alpha, \alpha - \mu \rangle}{\|\alpha - \mu\|^2} (\alpha - \mu)$  for any  $\mu \in W$  s.t.

$$\text{Then, } 0 \leq \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} - 2 \cdot \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} = - \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2}$$

And the inequality holds iff  $\langle \beta - \alpha, \alpha - \mu \rangle = 0$ , and all non zero vectors in  $W$  can be represented by  $\alpha - \mu$ , thus proved.

Conversely, if  $\beta - \alpha$  is orthogonal to every vector in  $W$ , then  $\operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle = 0$ , thus  $\|\beta - \gamma\|^2 - \|\beta - \alpha\|^2 = \|\alpha - \gamma\|^2 \geq 0$ .  $\square$

Thm. If b.app to  $\beta \in V$  by  $W$  exists, then it is unique.

Proof. Suppose that  $\alpha_1, \alpha_2 \in W$  satisfies for all  $\gamma \in W$ ,

$$\langle \beta - \alpha_1, \gamma \rangle = \langle \beta - \alpha_2, \gamma \rangle = 0.$$

Then, take  $\gamma = \alpha_1 - \alpha_2$ , so  $\langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle = 0$ .  $\square$

Def. For all  $\beta \in V$ , there is b.app  $\alpha \in W$ , then the map

$$P: V \rightarrow V: \beta \mapsto \alpha$$

is called an Orthogonal Projection on  $W$ .

**Thm.** Let  $W$  be a fin. dim subspace of inn. prod space  $V$ .

Then, the Unique orthogonal projection on  $W$  exists. Further,

$P$  is idempotent linear operator with  $\text{Im } P = W$ ,  $\text{Ker } P = W^\perp$ .

**Proof.** Let  $\{\alpha_1, \dots, \alpha_n\}$  be an orthonormal basis for  $W$ . Then, a map

$$P: V \rightarrow V: \beta \mapsto \sum_{i=1}^n \langle \beta, \alpha_i \rangle \cdot \alpha_i$$

1). It maps  $\beta$  to orthogonal projection by  $W$ .

For each  $1 \leq i \leq n$ ,

$$\left\langle \beta - \sum_{i=1}^n \langle \beta, \alpha_i \rangle \alpha_i, \alpha_j \right\rangle = \langle \beta, \alpha_j \rangle - \langle \beta, \alpha_j \rangle = 0.$$

So,  $\beta - P(\beta) \in W^\perp$  for all  $\beta \in V$ . (And, such a vector is unique, thus well-defined)

2). It is linear.

Let  $\beta, \gamma \in V$  and  $c \in F$  be given. Then,

$$P(c\beta + \gamma) = \sum \langle c\beta + \gamma, \alpha_i \rangle \cdot \alpha_i = c \cdot \sum \langle \beta, \alpha_i \rangle \cdot \alpha_i + \sum \langle \gamma, \alpha_i \rangle \cdot \alpha_i = cP(\beta) + P(\gamma).$$

3). It is idempotent.

Given  $\beta \in V$ ,  $P(\beta) \in W$ . So,  $P^2(\beta) = P(\beta)$ .  $\perp$  projection

4).  $\text{Im } P = W$ ,  $\text{Ker } P = W^\perp$ .

$\text{Im } P \subseteq W$  is clear, and for any  $\gamma \in W$ ,  $P(\gamma) = \gamma \in \text{Im } P$ . And,

$$\beta \in \text{Ker } P \Leftrightarrow P(\beta) = 0 \Leftrightarrow \langle \beta, \alpha_i \rangle = 0 \text{ for all } 1 \leq i \leq n \Leftrightarrow \beta \in W^\perp.$$

5). It is Hermitian.

Since  $\text{Im } P \oplus \text{Ker } P = V$ , let  $x_1, y_1 \in \text{Im } P$  and  $x_2, y_2 \in \text{Ker } P$  be given. Then

$$\langle P(x_1 + x_2), y_1 + y_2 \rangle = \langle x_1, y_1 + y_2 \rangle = \langle x_1, y_1 \rangle = \langle x_1 + x_2, y_1 \rangle = \langle x_1 + x_2, P(y_1 + y_2) \rangle.$$

□

**Corollary.** Let  $W$  be a fin. dim subspace of inn. prod space  $V$ ,

and let  $P$  be an orthogonal projection on  $W$ . Then,  $I - P$  is an orthogonal projection on  $W^\perp$ .

**Proof.** Given  $\beta \in V$ ,  $\langle \beta - P(\beta), \mu \rangle = 0$  for all  $\mu \in W$ , or  $\beta - P(\beta) \in W^\perp$ . So  $\text{Im}(I - P) = W^\perp$ .

And, given  $\gamma \in W^\perp$ ,  $\beta - \gamma = P(\beta) + (\beta - P(\beta) - \gamma)$ , therefore

$$\|\beta - \gamma\|^2 = \|P(\beta)\|^2 + \|\beta - P(\beta) - \gamma\|^2 \geq \|P(\beta)\|^2 = \|\beta - (\beta - P(\beta))\|^2 \text{ for all } W^\perp,$$

so  $\beta - P(\beta)$  is the best approximation to  $\beta$  by vectors in  $W^\perp$ .

Note: let  $E: V \rightarrow V$  be a projection. In general,  $E$  can not be determined by it's image.  
 For example, let  $V = \mathbb{R}^3$ ,  $W_1 = \text{Span}\{(1, 0, 0)\}$ ,  $W_2 = \text{Span}\{(0, 1, 0), (0, 0, 1)\}$ ,  $W_3 = \text{Span}\{(0, 1, 0), (1, 1, 1)\}$ .  
 Then,  $V = W_1 \oplus W_2 = W_1 \oplus W_3$ , but  $W_2 \neq W_3$ . Moreover,  
 $E: V \rightarrow V: (x_1, x_2, x_3) \mapsto (x_1, 0, 0)$ ,  $E': V \rightarrow V: (x_1, x_2, x_3) \mapsto (x_1 - x_3, 0, 0)$   
 Both  $E$  and  $E'$  are projections with  $\text{Im } E = \text{Im } E' = \text{Span}\{(1, 0, 0)\}$ , but  $E \neq E'$ ,  
 $\text{Ker } E = W_2$  and  $\text{Ker } E' = W_3$ .

Corollary. Let  $\{\alpha_i, \dots, \alpha_n\}$  be an orthogonal set of non-zero vectors in inner prod space  $V$ . Then,  
 for all  $\beta \in V$ ,  $\sum_{i=1}^n \frac{|\langle \beta, \alpha_i \rangle|^2}{\|\alpha_i\|^2} \leq \|\beta\|^2$ , the equality holds iff  $\beta = \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$ .

Proof. Take  $\gamma = \sum [\langle \beta, \alpha_i \rangle / \|\alpha_i\|^2] \cdot \alpha_i$ . Then,  $\langle \beta - \gamma, \gamma \rangle = 0$ . So  
 $\|\beta\|^2 = \|\gamma\|^2 + \|\beta - \gamma\|^2 \geq \|\gamma\|^2$ .

Thm. Let  $W$  be a finite-dimensional subspace of an inn. prod space  $V$ .

If  $P: V \rightarrow V$  is an orthogonal projection on  $W$ , that is,

$$\text{for any } \beta \in V, \quad \|\beta - P(\beta)\| \leq \|\beta - \gamma\| \text{ for all } \gamma \in W \quad \dots (*)$$

then,  $P$  is idempotent hermitian linear map s.t.  $\text{Im } P = W$ ,  $\ker P = (\text{Im } P)^\perp$ ,  $V = \text{Im } P \oplus \ker P$

Proof. Since  $P(\beta) \in W$ ,  $P(P(\beta)) = P(\beta)$ , for all  $\beta \in V$ , so  $P^2 = P$ , idempotent.

To show the linearity, let  $\alpha, \beta \in V$  and  $c \in F$  be given. Since  $\alpha - P(\alpha), \beta - P(\beta) \in W^\perp$ , so

$$c \cdot (\alpha - P(\alpha)) + (\beta - P(\beta)) = (c\alpha + \beta) - (c \cdot P(\alpha) + P(\beta)) \in W^\perp.$$

So,  $cP(\alpha) + P(\beta) = P(c\alpha + \beta)$ , linearity is obtained

To show the direct sum, given  $\beta \in V$ ,  $\beta = P(\beta) + (\beta - P(\beta))$ , and since  $\beta - P(\beta) \in W^\perp$ ,  $V = \text{Im } P + W^\perp$  is obtained. And,

$$\alpha \in W^\perp \Leftrightarrow P(\alpha) = 0, \text{ because}$$

$$\alpha \in W^\perp \Leftrightarrow \forall \mu \in W, \langle \alpha, \mu \rangle = 0 \Leftrightarrow P(\alpha) = 0. \text{ So, } V = \text{Im } P + \ker P. \text{ And, } \text{Im } P \cap \ker P = \{0\},$$
$$\langle \alpha - 0, \mu \rangle = 0.$$

because  $P(0) = 0$  and  $\alpha = P(\alpha)$  implies  $\alpha = 0$ .

